

THE THREE TRIBES OF EMBODIED INTELLIGENCE

Bridging the Disciplinary Divides Between Computer Science, Embedded Systems, and Mechanical Robotics

TRIBE 1: COMPUTER SCIENCE / AI

Maya's Perspective

- Loss functions · Transformers · Latent world models
- Python abstractions · Generative policies · Tokenization

TRIBE 2: EMBEDDED SYSTEMS / ECE

Carlos's Perspective

- Microsecond jitter · DMA memory buses · Interrupt service routines
- Static SRAM (malloc=0) · Watchdogs · AXI QoS priorities

PHYSICAL
AI

TRIBE 3: ROBOTICS & MECHANICS

Alex's Perspective

- Kinetic momentum · Friction cones · Contact compliance
- Stopping distance d_{stop} · Gearbox backlash · Dynamic stability